

Muhammad Mostafa Saadawy Amin

Junior Data Engineer

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PROFESSIONAL SUMMARY

Junior Data Engineer and final-year Computers & AI student with hands-on experience building scalable batch and streaming ETL pipelines in Python, Spark, Kafka, and Airflow.

Skilled in designing Star Schema data warehouses, improving query performance by 30–40%, and deploying analytics solutions on Microsoft Azure.

Strong problem-solving skills and analytical thinking developed through independent, end-to-end project work, paired with clear communication and interpersonal skills built through effective collaboration with cross-functional and academic teams. Focused on real-time data processing and CI/CD automation at scale.

PROFESSIONAL EXPERIENCE

Junior Data Engineer – Trainee (DEPI Scholarship) | Ministry of Communications – DEPI Program *Jun 2025 – Dec 2025*

- Built and maintained 10+ end-to-end batch and streaming ETL pipelines using Python, SQL, Apache Spark, Apache Kafka, and Apache Airflow for data ingestion and transformation.
- Designed 5+ data warehouse schemas (Star Schema) and optimized analytical queries, improving query performance by 30–40% while applying industry standards for data modeling.
- Processed and analyzed millions of records using Hadoop, Hive, and Spark for large-scale big data analytics, enabling scalable and efficient data processing workflows.
- Implemented cloud-based analytics solutions on Azure Synapse Analytics, Dedicated SQL Pools, and Azure Stream Analytics, supporting near real-time data processing.
- Applied CI/CD pipelines to automate deployment, orchestration, and monitoring using Git and GitHub Actions, reducing manual intervention by 40%.
- Worked with Azure Data Factory for data integration and pipeline orchestration across cloud data sources, coordinating closely with trainers and peers to troubleshoot issues.

Machine Learning Trainee | National Telecommunication Institute (NTI) *Aug 2025 – Sep 2025*

- Developed and evaluated machine learning models using Python, NumPy, Pandas, and Scikit-learn for real-world datasets.
- Applied data preprocessing, feature engineering, and model tuning to improve model performance and accuracy.
- Implemented supervised and unsupervised learning algorithms including regression, classification, and clustering.
- Built end-to-end ML workflows from data collection and cleaning to model training, evaluation, and deployment via REST APIs using Streamlit.

PROJECTS

IKEA Shopping – Admin Panel + Data Engineering & ML Pipeline *Apr 2026 – Jun 2026*

[GitHub](#) | [Live Demo](#)

- Extended a Java Swing admin panel (SQL Server, 10-table normalized schema, RBAC, SHA-256 hashing) with a full data engineering and ML layer, without modifying any existing Java code.
- Built a synthetic data generator producing 18 months of realistic sales data with monthly seasonality, weekly patterns, and growth trend using NumPy and pyodbc.
- Designed and implemented an ETL pipeline (Extract from SQL Server → Transform into a Star Schema → Load into SQLite warehouse) with Fact_Sales, Dim_Product, and Dim_Date tables.
- Trained an XGBoost forecasting model with lag features (lag-1, lag-7), 7-day rolling mean, and chronological train/test split; deployed a live Streamlit dashboard showing actual vs. forecasted daily sales per product, applying creative problem-solving to handle sparse and seasonal data.

Technologies: Java, Swing, SQL Server, Python, Pandas, NumPy, XGBoost, Streamlit, SQLite, ETL, Data Warehousing

Smart City – Real-Time Data Pipeline *Dec 2025 – Apr 2026*

[GitHub](#)

- Built an end-to-end real-time data pipeline simulating 5 smart city IoT sources (vehicles, GPS, traffic cameras, weather, emergencies) using Python producers and Apache Kafka.

- Processed concurrent Kafka streams with Apache Spark Structured Streaming, applying per-topic schemas, JSON deserialization, and watermarking to handle late-arriving events.
- Containerized the full distributed system (Zookeeper, Kafka broker, Spark master + 2 workers) using Docker Compose for one-command, reproducible deployment.
- Modeled data entities and relationships via ERD across 5 interconnected tables to support downstream analytics.

Technologies: Python, Apache Kafka, Apache Spark Structured Streaming, Docker, Docker Compose, Apache Parquet

Road Collisions Severity Prediction – ML Project

Sep 2025 – Nov 2025

[GitHub](#) | [Live Demo](#)

- Built an end-to-end ML system to predict road collision severity (92.6% accuracy) using real-world traffic accident data.
- Performed data cleaning, feature engineering, and EDA to identify key factors affecting accident severity.
- Trained and evaluated a CatBoost classifier, achieving a 0.92 weighted F1-score with perfect precision on Fatal and Serious collision classes.
- Deployed an interactive Streamlit web application for real-time severity predictions.

Technologies: Python, CatBoost, Scikit-learn, Pandas, Streamlit, Matplotlib, Seaborn

Spam Email Classifier – NLP Project

Aug 2025

[GitHub](#) | [Live Demo](#)

- Developed an end-to-end NLP pipeline for spam detection using tokenization, stopword removal, lemmatization, and TF-IDF vectorization.
- Trained and evaluated classification models using Scikit-learn; managed model persistence with Joblib.
- Deployed an interactive Streamlit dashboard for real-time predictions with probability breakdown.

Technologies: Python, NLTK, Scikit-learn, TF-IDF, Pandas, Streamlit, Git, GitHub

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, C#

Data Engineering: ETL Pipelines, Data Modeling, Data Warehousing, Star Schema, Batch & Streaming Processing

Big Data Analytics: Apache Spark, PySpark, Apache Kafka, Hadoop, Hive, Spark Structured Streaming

Orchestration: Apache Airflow, CI/CD Automation, GitHub Actions, dbt

Cloud (Azure): Azure Synapse Analytics, Azure Data Factory, Azure Databricks, Dedicated SQL Pools, Azure Stream Analytics

Databases: PostgreSQL, MySQL, Delta Lake, MongoDB (NoSQL)

DevOps & Tools: Docker, Docker Compose, Git, GitHub, REST APIs, Pipeline Monitoring, Industry Standards & Best Practices

ML Libraries: Scikit-learn, Pandas, NumPy, CatBoost, Matplotlib, Seaborn, NLTK, Streamlit

Human Languages: Arabic (Native), English (Professional Working Proficiency)

EDUCATION

Bachelor of Computers and Artificial Intelligence

Expected: 2027

Minia National University, Egypt

Relevant Coursework: Data Structures, Data Mining, Algorithms, Database Systems, OOP, Machine Learning

LEADERSHIP & ACTIVITIES

- Mentor at ICPC MNU Team — guided competitive programming students in problem-solving and algorithms, strengthening peer interpersonal and coaching skills.
- Mentor at MENTOR Network — supported junior students in career development and technical growth.
- Admin at Student Activity — managed organizational and academic community events, collaborating with student committees.